

Previews

An exception to the rule? Regeneration of the injured spinal cord in the spiny mouse

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The capacity for long-distance axon regeneration and functional recovery after spinal cord injury in the adult has long been thought to be a unique feature of certain non-mammalian vertebrates. However, in this issue of *Developmental Cell*, Nogueira-Rodrigues et al. report an astonishingly high regenerative ability in the spiny mouse.

Long-distance axon regeneration and functional recovery after injury is very limited in the mature central nervous system (CNS) of mammals, including humans (Bradbury and Burnside, 2019). So far, only non-mammalian vertebrate species, including fish and urodele amphibians, have been shown to naturally regrow axonal connections and recover motor function after spinal cord injury (SCI) (Cigliola et al., 2020). One of the reasons for regenerative failure in mammals is an inhibitory extracellular matrix (ECM) in the injury site, which is associated with scarring (Burnside et al., 2018). In fish, on the other hand, the ECM is growth-promoting, owing to a more favorable composition of matrix molecules (Tsata et al., 2021; Wehner et al., 2017). This suggests that the different responses to CNS lesions among vertebrate species relate to differences in the molecular composition of the injury-induced ECM.

To find a mammal that could show better capacity for spinal cord regeneration, in this issue of *Developmental Cell*, Nogueira-Rodrigues et al. turned to the spiny mouse (*Acomys cahirinus*) (Nogueira-Rodrigues et al., 2021). *Acomys* is a member of the *Muridae* family that also includes the widely used experimental model organism *Mus musculus* and is well known for showing a reduced scarring response after injury of the skin, ear, muscle, kidney, spinal cord, and heart (Gaire et al., 2021). Indeed, the authors report that *Acomys* exhibits considerable axonal regrowth and functional recovery, including bladder control, after SCI (Figure 1). They also

present evidence that the lesion ECM composition is a critical determinant of regenerative success. This establishes *Acomys* as a mammalian species capable of natural adult CNS regeneration.

To determine the regenerative capacity of the CNS of adult *Acomys*, Nogueira-Rodrigues et al. assessed axonal regrowth by immunolabelling after spinal cord transection at the eighth thoracic vertebra. A high number of tracts penetrating and spanning the transection site were present in *Acomys* at 8 weeks after injury (WPI). Among these were descending serotonergic fibers from the motor raphespinal tract, which interestingly also exhibits some growth capacity in the CNS of other *Muridae* family members (Jin et al., 2016) and ascending (sensory) pathways. Tracing experiments revealed bilateral axon sprouting of the voluntary movement-controlling corticospinal tract caudal to the lesion site at 12 WPI, which does not occur in *Mus*. Regenerated corticospinal fibers were decorated with a pre-synaptic marker, suggesting re-establishment of synaptic connectivity. Furthermore, in half of the injured *Acomys*, spinal cords conducted lateral funiculus-stimulated compound action potentials across the lesion site at 12 WPI. Nogueira-Rodrigues et al. next assessed functional recovery with the standardized Basso mouse scale (BMS) score for locomotion (Basso et al., 2006). Within two months, *Acomys* re-established the initial weight support with occasional plantar stepping, corresponding to an average BMS score of 4 (uninjured animals score a 9 on that scale). Notably,

about half of the animals achieved full weight-bearing hindlimb standing and presented restored pattern of fore and hind limb coordination. Even more remarkable is the apparent full recovery of bladder control, a major concern for SCI patients. By using a urine spot assay, injured *Acomys* were indistinguishable from sham controls within weeks after SCI, suggesting recovery of normal bladder control. The extent of functional recovery presented by *Acomys*, although incomplete based on the BMS score, far exceeds that reported for any other adult mammal after a complete SCI. Examining the lesion site, the authors found evidence for reduced fibrous scarring in *Acomys*, providing a possible explanation for the enhanced regenerative capacity. Comparative transcriptional profiling of the lesion site between *Acomys* and *Mus* revealed differences in the levels of enzymes of the biosynthetic pathways of keratan sulfate (KS), heparan sulfate, and chondroitin sulfate (CS) glycosaminoglycans; polysaccharide chains found attached to proteoglycan core proteins. This suggested differences in the composition of the injury ECM between *Acomys* and *Mus*. $\beta 3gnt7$, a key enzyme of KS glycosaminoglycan biosynthesis, was among the highly upregulated enzymes in *Acomys*. Expression of $\beta 3gnt7$ coincided with an increased anti-KS immunoreactivity in *Acomys* but not *Mus*, suggesting permissive axon growth, or even promoting properties of KS glycosaminoglycans. Indeed, $\beta 3gnt7$ -expressing CHO cells provided a neurite growth-stimulating



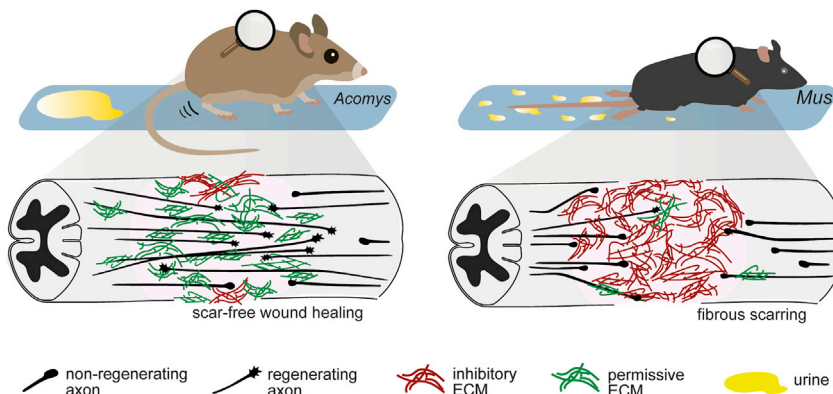


Figure 1. Reduced fibrous scarring facilitates axon regeneration and functional recovery after SCI in the adult spiny mouse

After SCI, *Acomys* (left), regains bladder control and some motor function, possibly due to an axon growth-permissive ECM in the injury site. *Mus* (right) retains permanent deficits because the severed axon cannot penetrate the inhibitory ECM.

substrate *in vitro*, implicating a KS proteoglycan (KSPG)-rich ECM as supportive for axon growth. Further support could come from a growth-conductive CS glycosaminoglycans pattern, suggested by an altered expression of CS-sulfotransferase enzymes, which remains to be fully analyzed.

The work by Nogueira-Rodrigues et al. is exciting because it presents the first in-depth report on a mammalian species with an elevated regenerative capacity for the adult CNS. It also supports the emerging view that the balance of growth promoting and inhibitory components of the injury ECM ultimately determines the ability of axons to bridge the lesion site after SCI. Several questions arise from their discovery. First, which are the regenerating pathways that enable full recovery of bladder control yet imperfect locomotor recovery in *Acomys*? Systematic mapping using neuronal tract-tracing would be very informative. This could also reveal which regenerated axon tracts correlate with the best regeneration of motor function, which is highly variable. Second, would experimentally increasing $\beta 3\text{gnt}7$ levels and, thus, KS glycosaminoglycan abundance enhance permissiveness of the injury ECM in *Mus*? This is particularly

interesting because KSPGs are historically considered detrimental to neurite growth (Imagama et al., 2011). Similarly, does depletion of KS and/or of the potentially conductive CS glycosaminoglycan signature alter the permissiveness of the injury ECM in *Acomys*? Finally, is there a common signature of ECM components among different vertebrates with high regenerative capacity? A multiple cross-species comparison would shed light on which ECM components permit and which inhibit regeneration after CNS injury. Nogueira-Rodrigues et al. have greatly expanded the toolbox of model organisms that offer a blueprint to address one of the last major questions in the field of regenerative biology: Why are some vertebrates able to regenerate their spinal cord after injury? Elucidating the differences and communalities in regenerating, partly, and non-regenerating species could provide strategies to modulate the ECM composition of the lesion site and tip the balance toward successful spinal cord repair in all mammals.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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